



# Physically Unclonable Function (PUF)

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Group 2

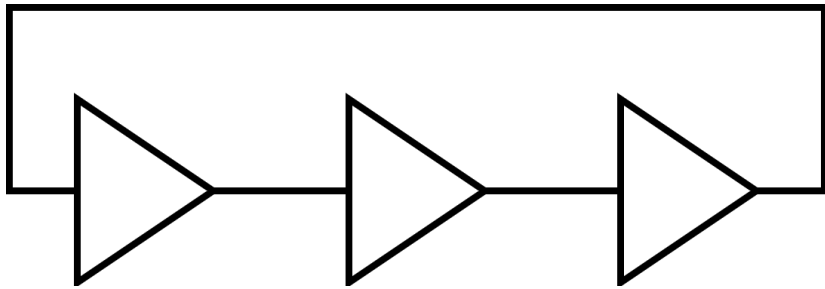
Alex Knowlton, Avinash Gupta, Bao Vo, Ali Hammoud, Yesnas Gowda, Rahul N.

# PUF Intro

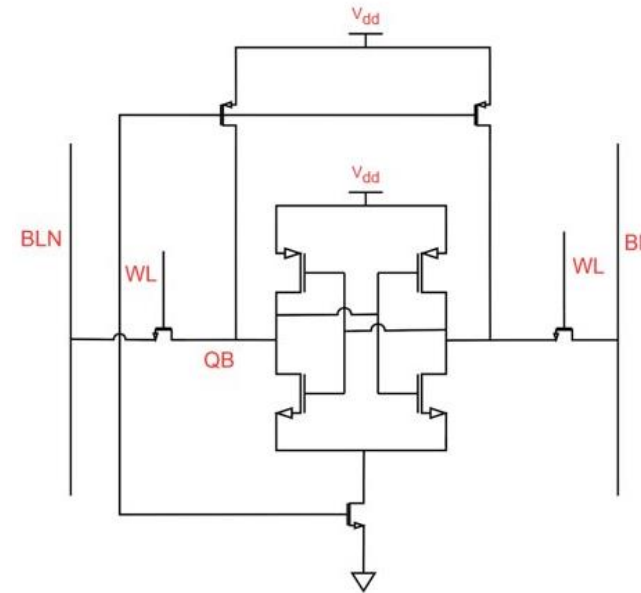
- Physically Unclonable Function
- Process variation to generate a unique number
- NOT a random number generator
- PUF value is constant for a single chip
- All PUFs design at unstable equilibrium
- Respond to challenge bits with different outcomes

# Architecture Comparison

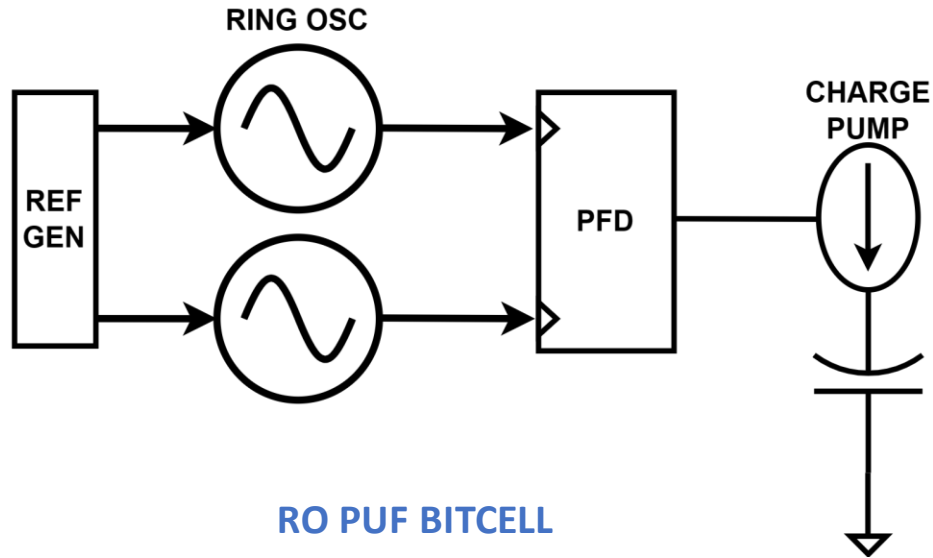
# Ring Oscillator



## SRAM



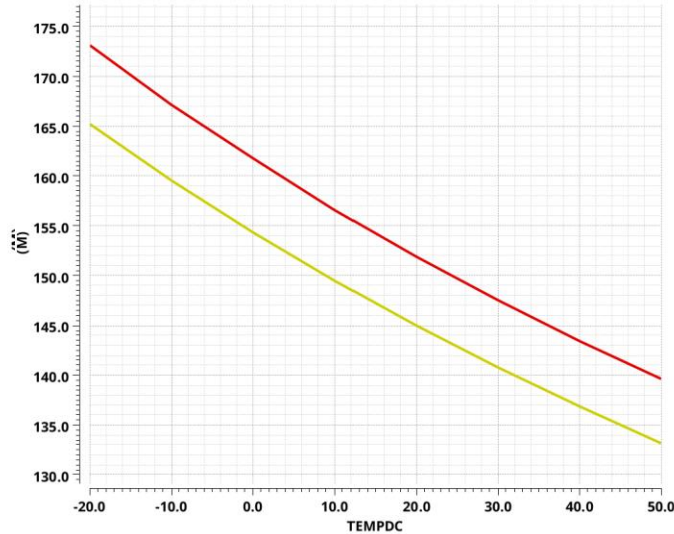
# RO PUF: System Overview



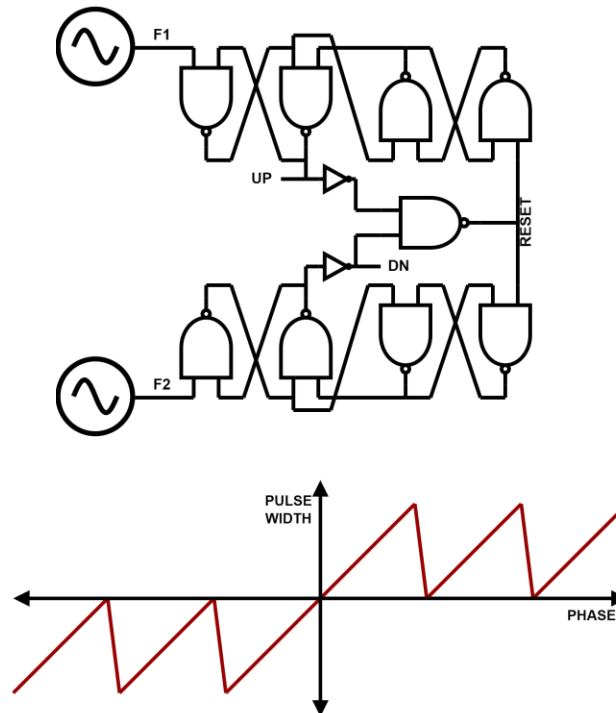
- To reuse the charge pump, PFD and capacitor we need to add a decoder between the ring oscillators and the PFD. (synthesized in verilog).
- All possible combinations of ring oscillators (16) are paired result in 120 PUF bits.
- 'Challenges' are used to select groups of 16.

# RO PUF – Temperature Invariance

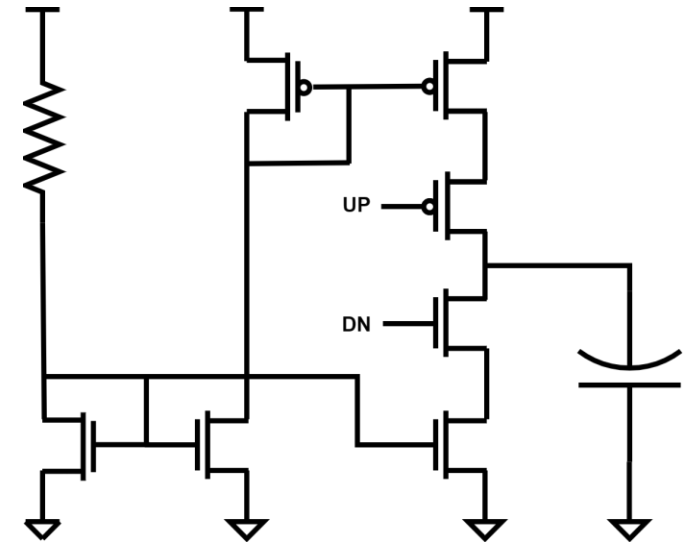
- First Order Temperature Invariance:
  - Compare with another ring oscillator frequency (similar temperature dependence)
- Convert Pulse at output of PFD into 0/1.
  - Use a charge Pump



## PHASE FREQUENCY DETECTOR

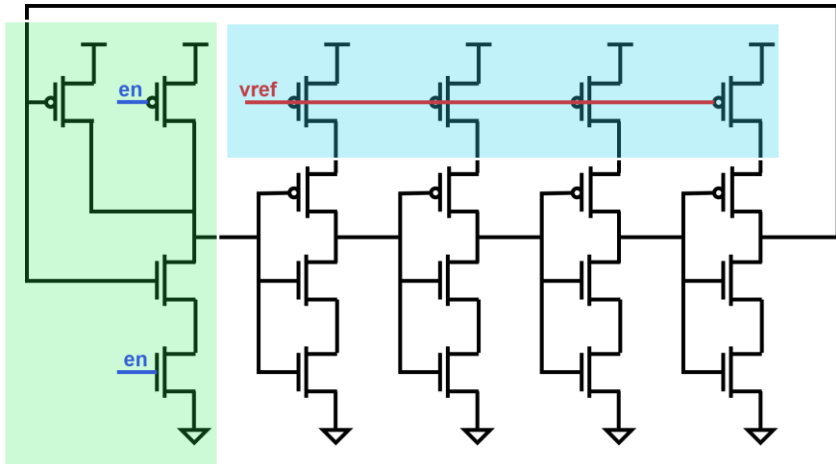
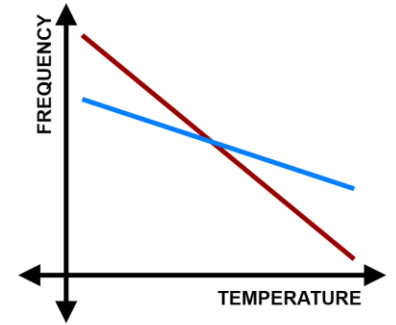


## CHARGE PUMP

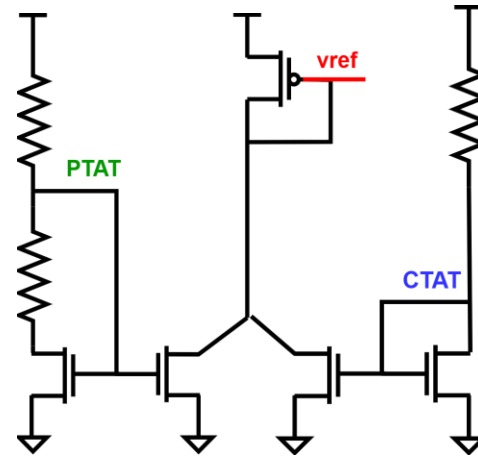


# RO PUF – Temperature Invariance

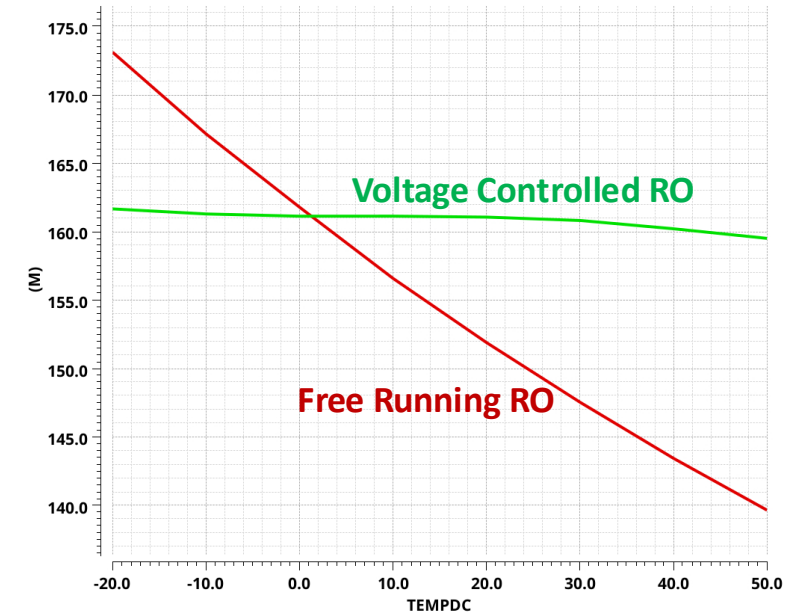
- Second Order Temperature Invariance – Unequal Frequency Slope.
  - The generated PUF bit might flip with temperature.



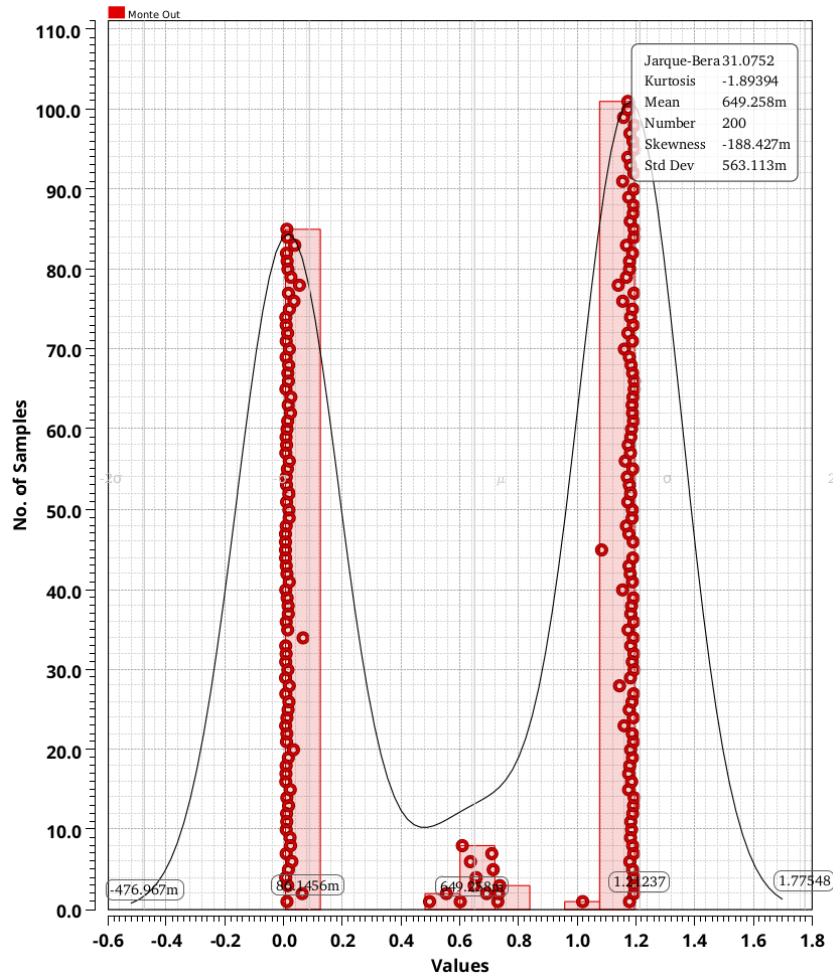
RING OSCILLATOR



VOLTAGE REFERENCE



# RO PUF: Results and Issues.



Monte Carlo showing ~95% Yield

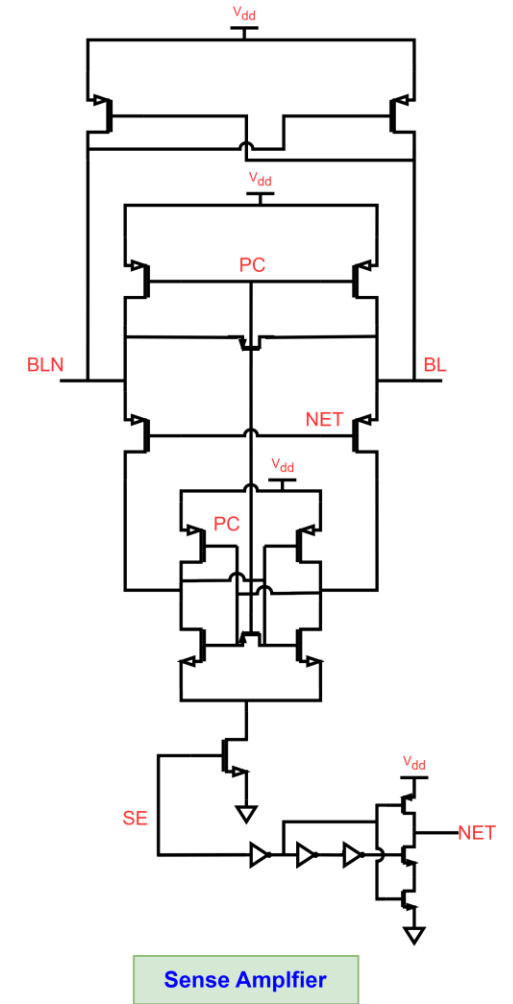
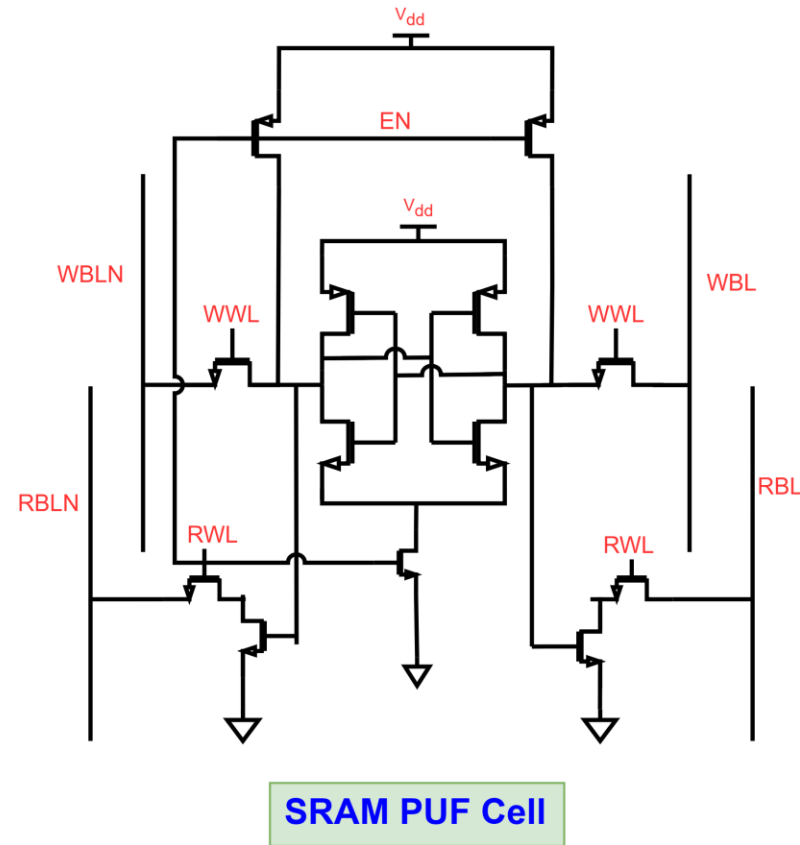
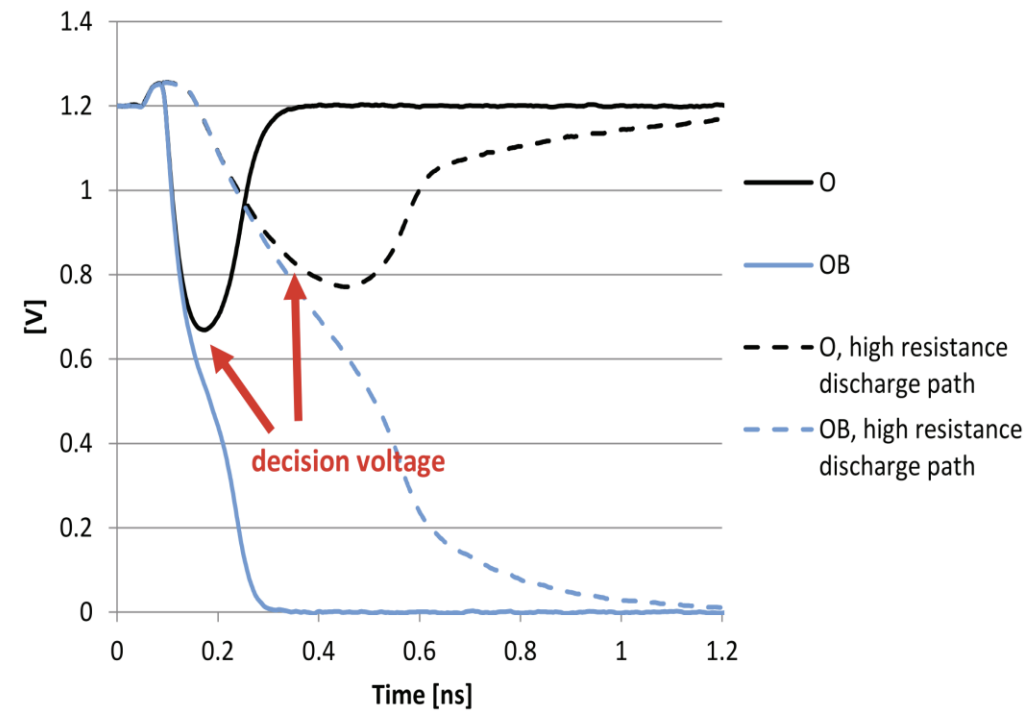
## PFD Issue – Long generation time

- 'Setup' constraints of phase frequency detector for very small phase differences causes pulses in the opposite direction.
- To minimize the effect of incorrect charging/discharging, one of these three solutions can work.
  - Reduce RO Frequency.
  - Reduce Charge Pump Current
  - Reduce Output Capacitor.

## Charging Time

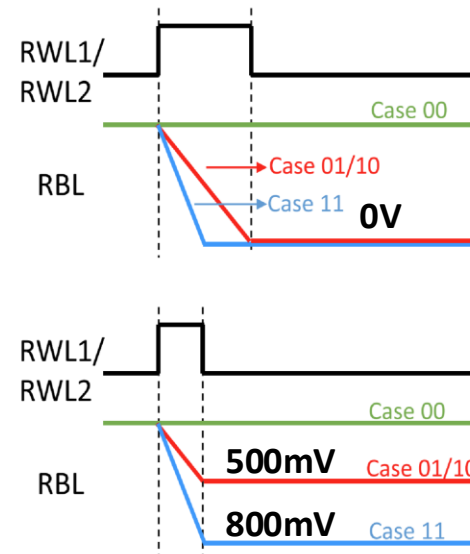
- For very small frequency variations, the charging time can be very long

# SRAM PUF Design

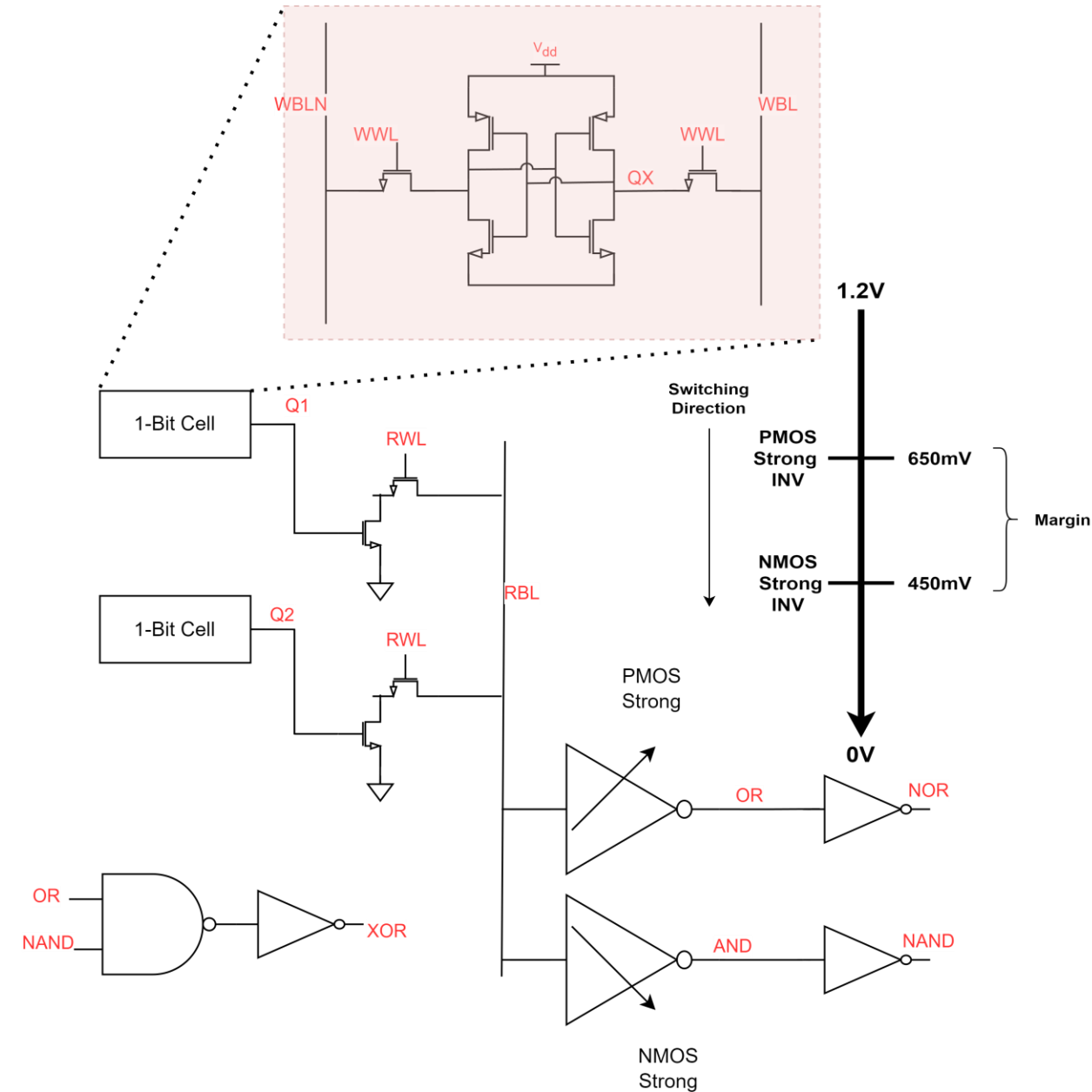


# SRAM CIM Design

- Need huge inverters as we need to skew the inverters to increase the margin.
- Logic fails in MC sims.
- Need timing control for RWL.



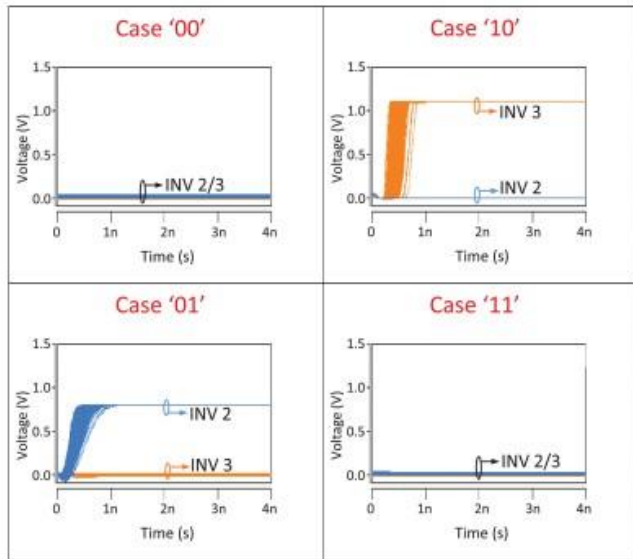
| Q <sub>1</sub> | Q <sub>2</sub> | RBL   | NAND | NOR | XOR |
|----------------|----------------|-------|------|-----|-----|
| 0              | 0              | 1.2 V | 1    | 1   | 0   |
| 0              | 1              | 500mV | 1    | 0   | 1   |
| 1              | 0              | 500mV | 1    | 0   | 1   |
| 1              | 1              | 0 V   | 0    | 0   | 0   |



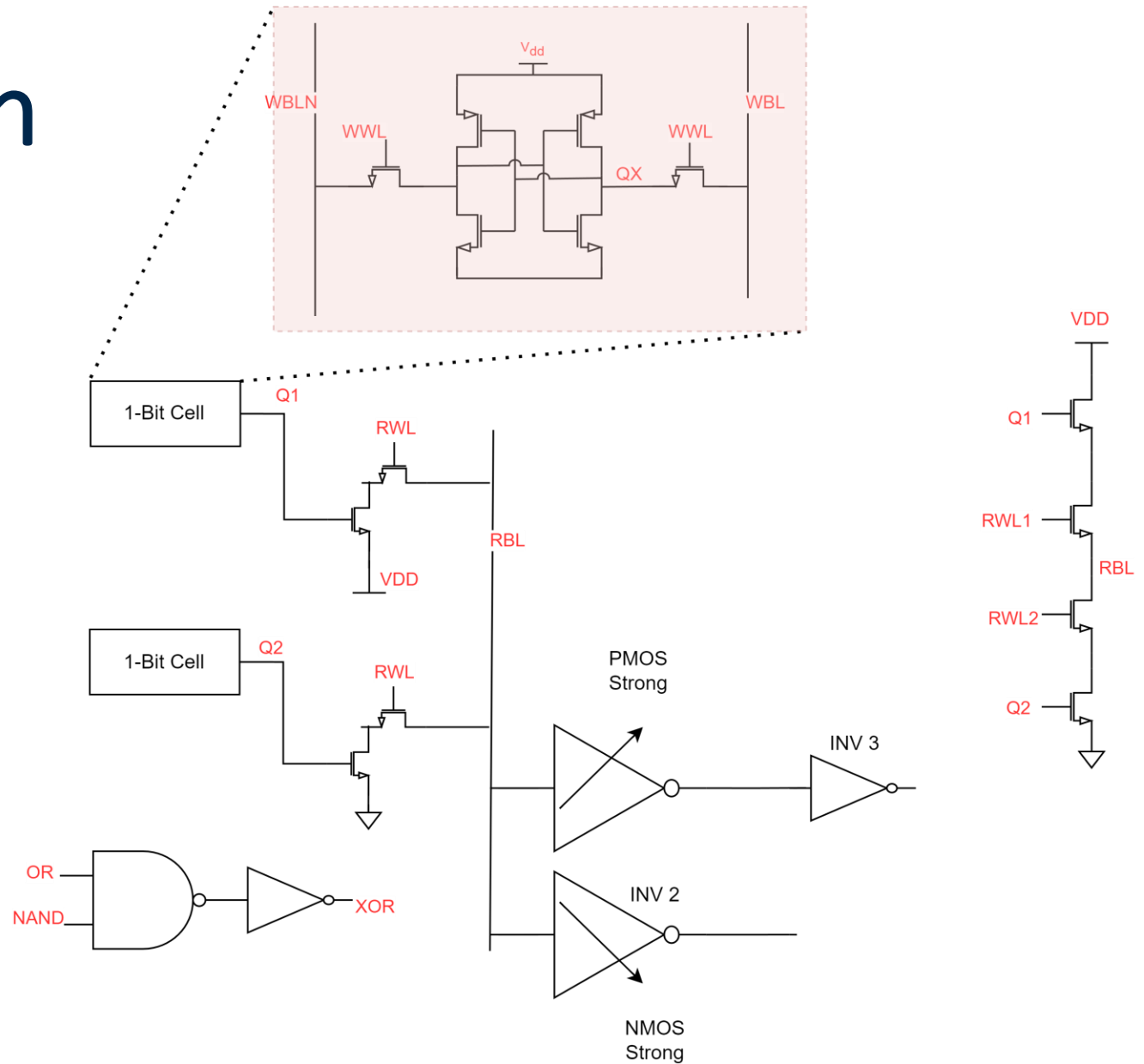
Amogh Agarwal et al., IEEE TCAS 1(2018), Vol. 65, Issue: 12

# SRAM CIM Design

- Need huge inverters as we need to skew the inverters to increase the margin.
- Logic fails in MC sims.
- No need of timing control for RWL.



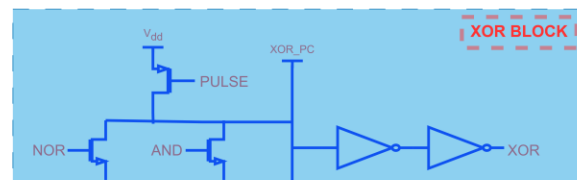
Amogh Agarwal et al., IEEE TCAS 1(2018), Vol. 65, Issue: 12



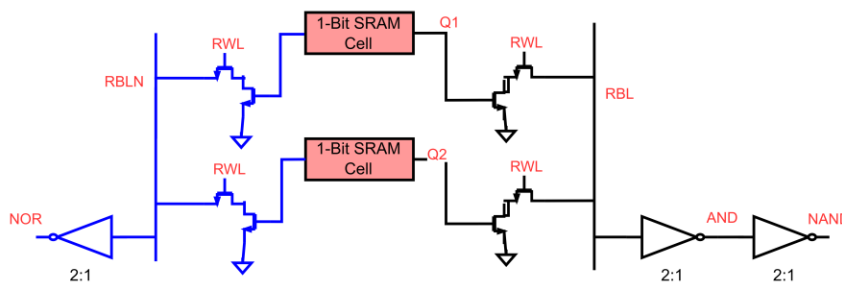
# SRAM CIM Design

## Improvements

|                       |
|-----------------------|
| Area Reduced          |
| Robust                |
| No Extra Bias Circuit |
| No Static Logics      |
| Improved Speed        |



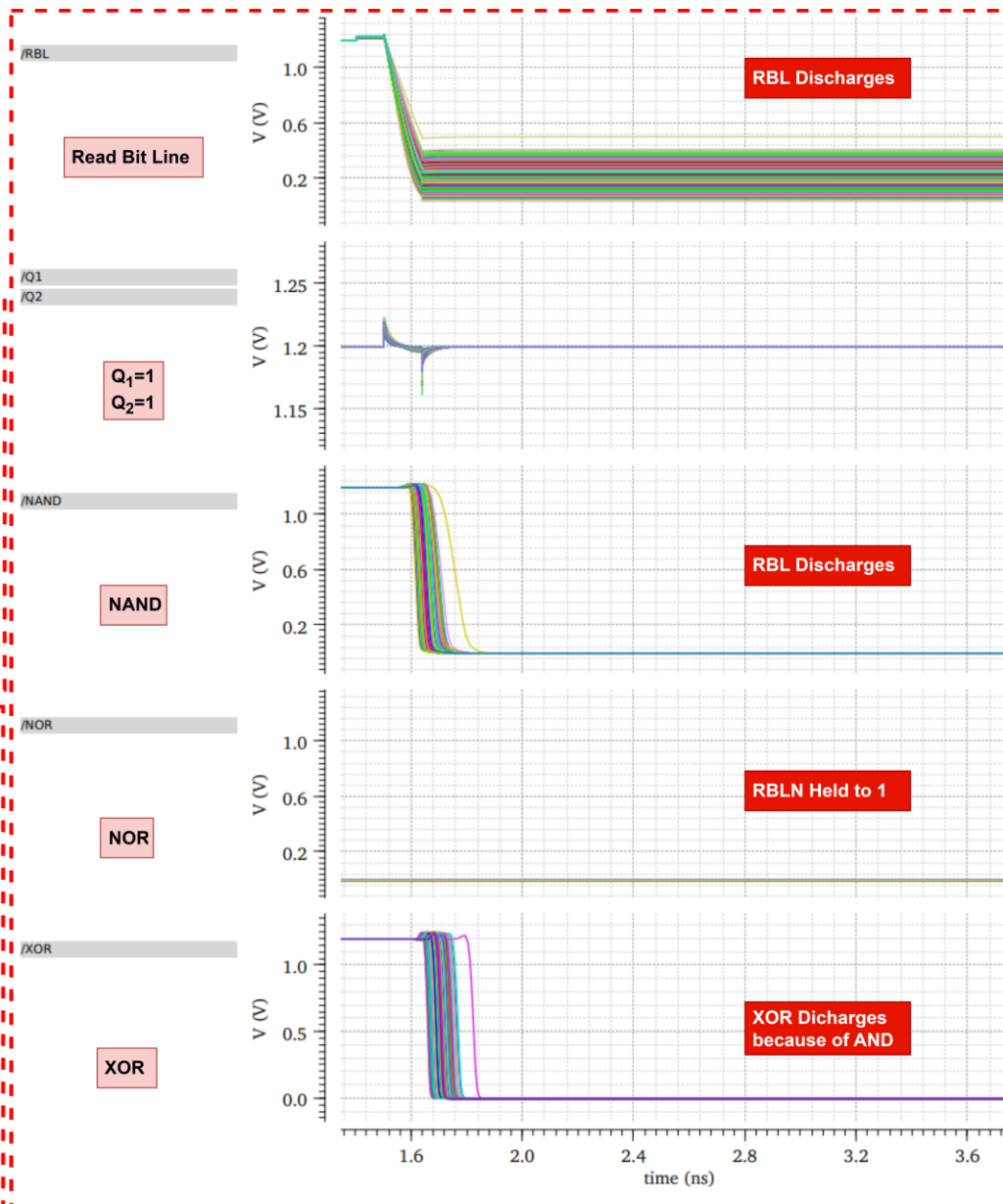
| Q <sub>1</sub> | Q <sub>2</sub> | AND | NOR | XOR_PC    | XOR |
|----------------|----------------|-----|-----|-----------|-----|
| 0              | 0              | 0   | 1   | Discharge | 0   |
| 0              | 1              | 0   | 0   | OFF       | 1   |
| 1              | 0              | 0   | 0   | OFF       | 1   |
| 1              | 1              | 1   | 0   | Discharge | 0   |



| Q <sub>1</sub> | Q <sub>2</sub> | V <sub>clip</sub> | NAND |
|----------------|----------------|-------------------|------|
| 0              | 0              | 1.2 V             | 1    |
| 0              | 1              | 0.8 V             | 1    |
| 1              | 0              | 0.8 V             | 1    |
| 1              | 1              | 0 V               | 0    |

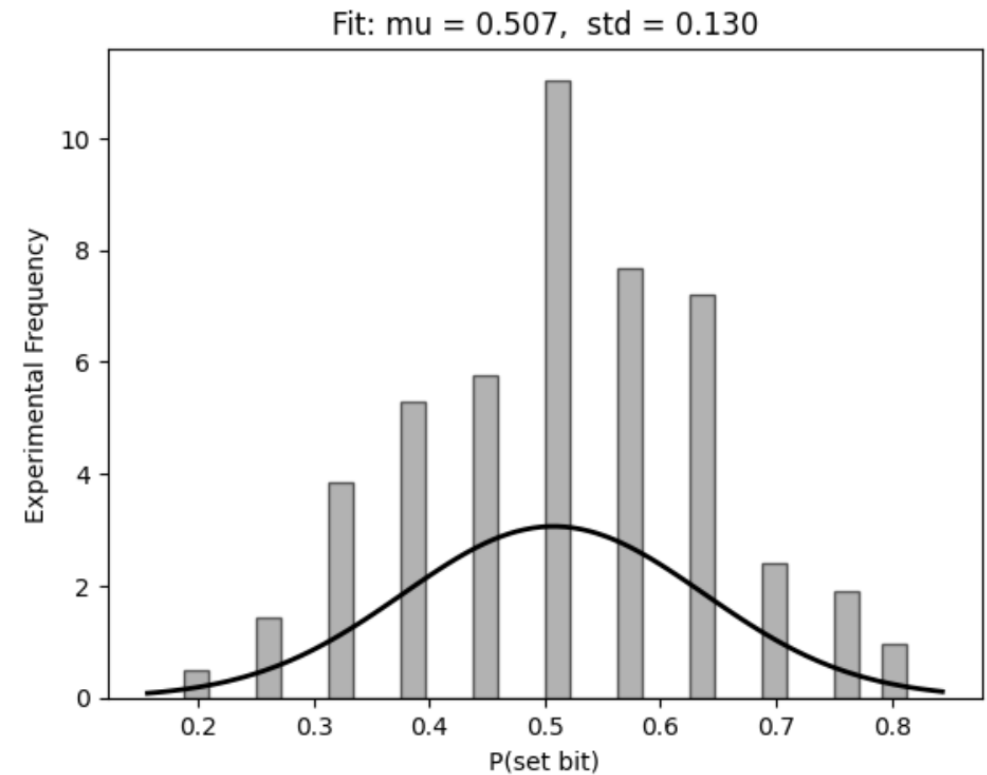


| Q <sub>1_B</sub> | Q <sub>2_B</sub> | V <sub>clip</sub> | NOR |
|------------------|------------------|-------------------|-----|
| 1                | 1                | 0 V               | 1   |
| 1                | 0                | 0.8 V             | 0   |
| 0                | 1                | 0.8 V             | 0   |
| 0                | 0                | 1.2 V             | 0   |



# SRAM Masking

- Difficult to deal with temp invariance
- Easiest way is to add cells
- More cells -> higher likelihood of stable output
- 16 cells and Monte Carlo give idea of yield



# Device Comparison

- Temp invariance
  - Bit Masking
- SRAM advantages: read compute store + encryption
- Area: wrapping layout

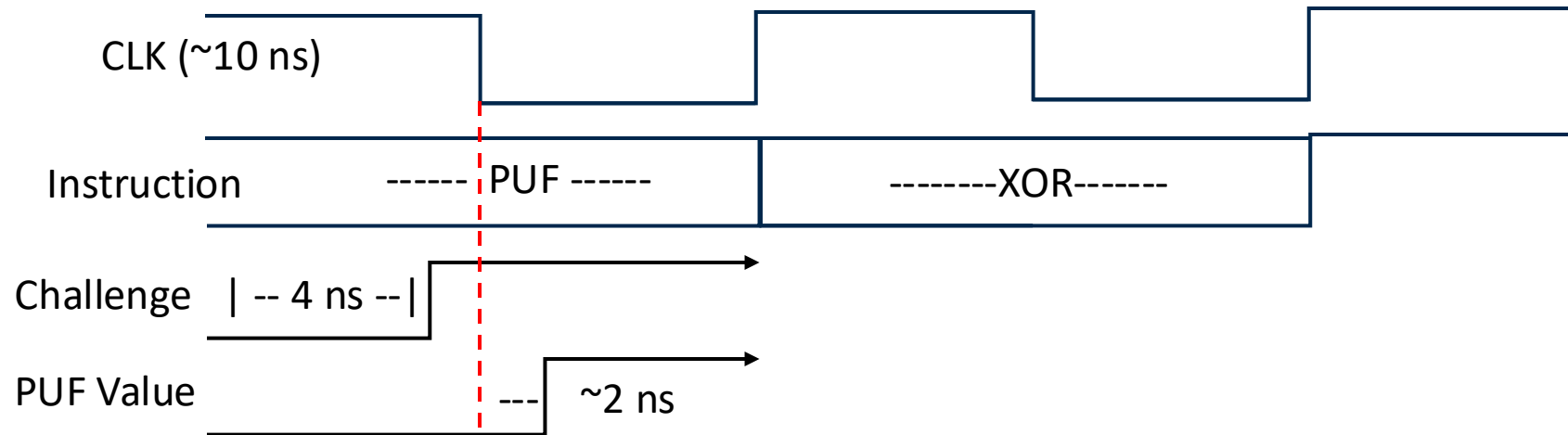
| FOM             | Ring Oscillator PUF | SRAM PUF |
|-----------------|---------------------|----------|
| Total Yield     | 95%                 | 11%**    |
| Area (um^2)     | ~4000               | 8110     |
| *Bernoulli(0.5) | yes                 | yes      |

\*Both models converged to no statistically significant difference relative to desired distribution

\*\*After bit masking

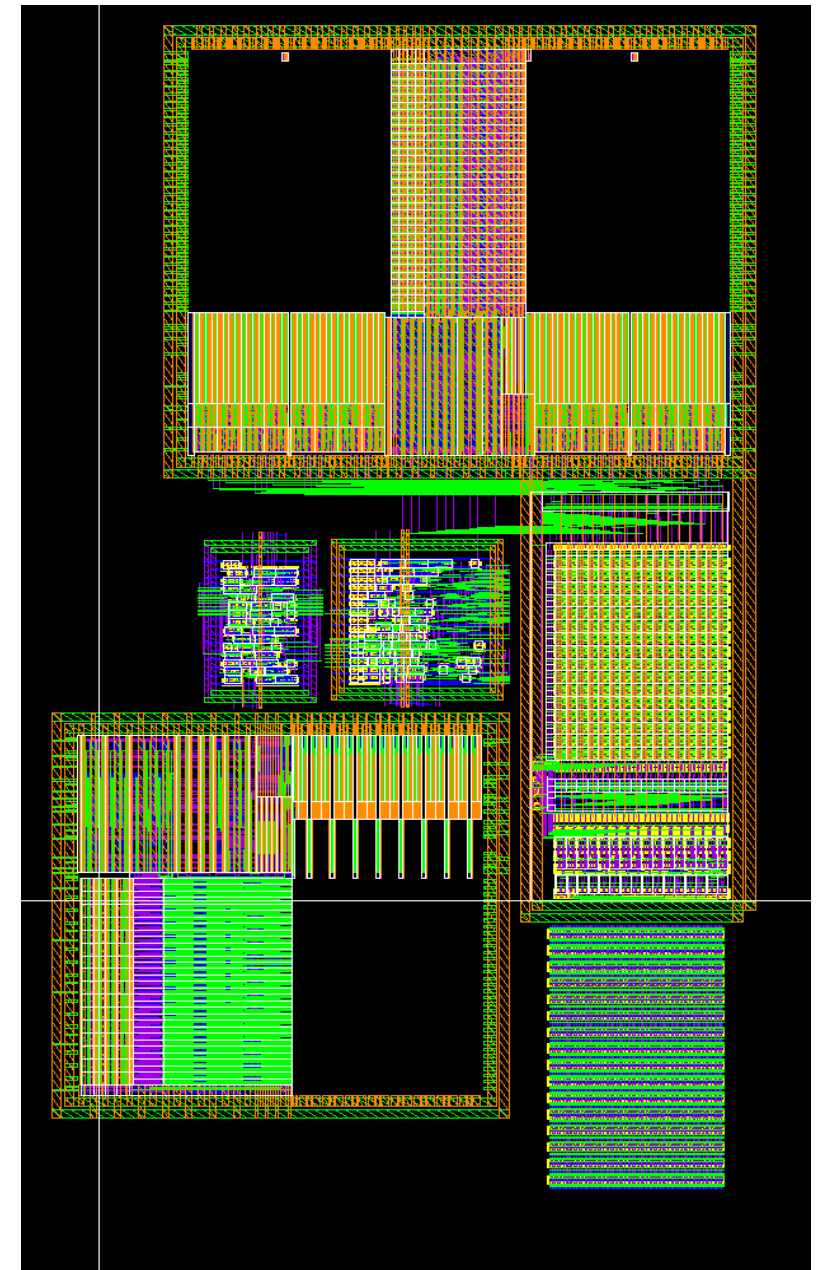
# PUF Instruction

- PUF Rdest, Challenge
- Latch @ negative clk edge
- Stores the result of challenging the PUF in R<sub>dest</sub>

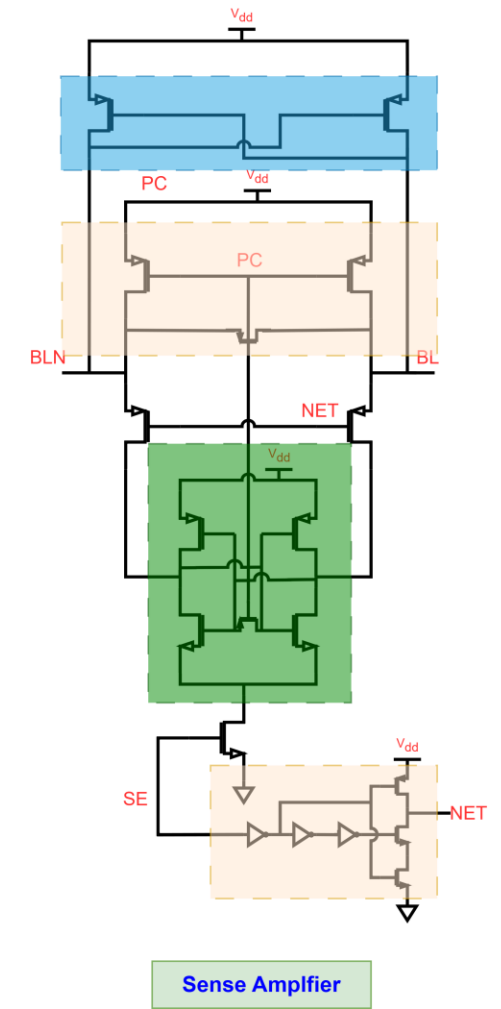


# PUF Integration

- PUF output MUXed with ALU output
- Pending toplevel integration
- More analog-heavy
- Lots of tradeoffs
- SRAM very useful for CIM but difficult as PUF
- PUF useful for IoT security
- Possible use for Encrypt-in-memory



# SRAM Sense Amplifier Design



# Appendix Slides (1/N) - PFD

